



Database systems

ITCS2322

Chapter 13 : Business Intelligence and Data Warehouses

اعداد : دكتور محمد عبد اللطيف راضي

Learning Objectives

- In this chapter, students will learn:
 - How business intelligence provides a comprehensive business decision support framework
 - About business intelligence architecture, its evolution, and reporting styles
 - About the relationship and differences between operational data and decision support data
 - What a data warehouse is and how to prepare data for one

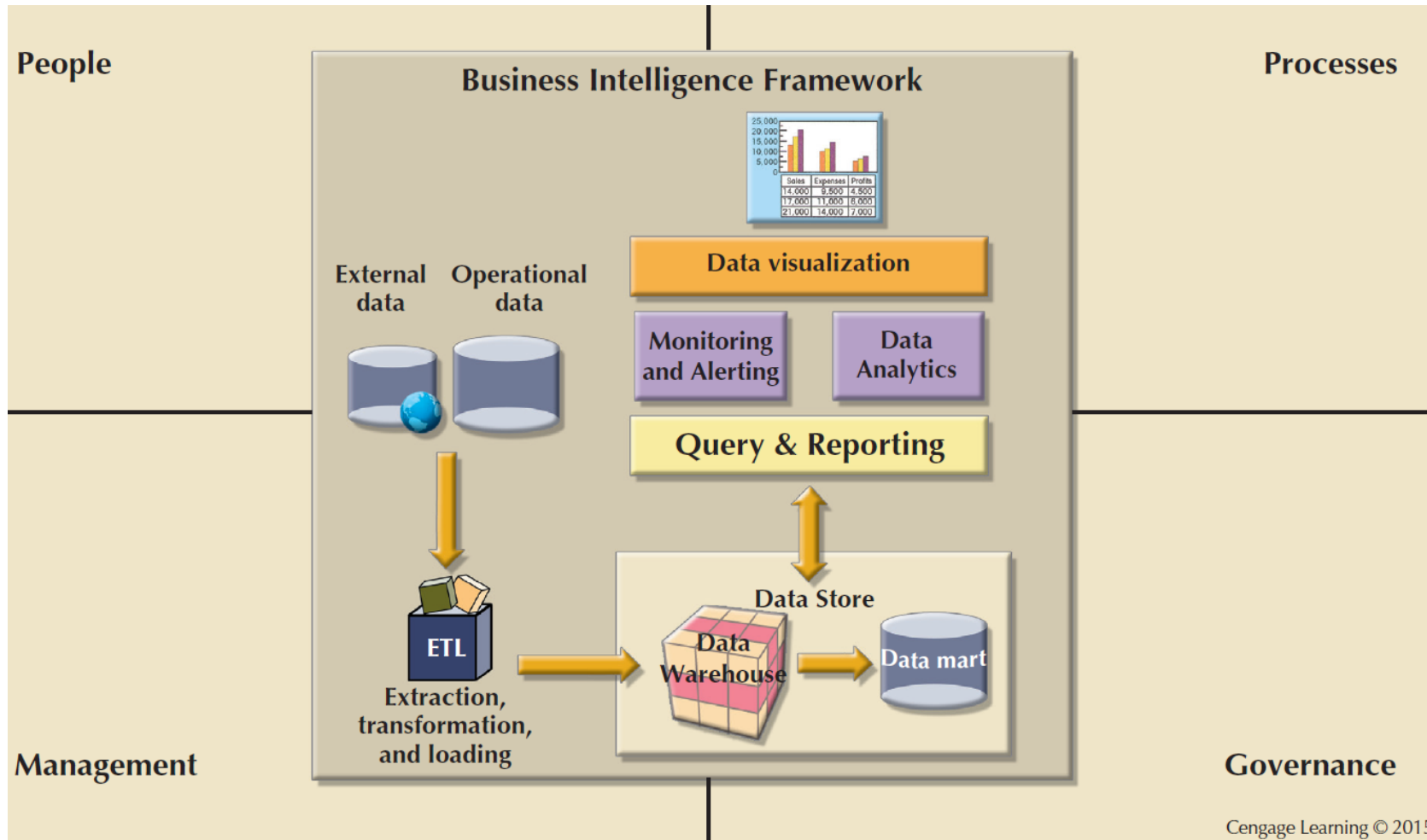
Learning Objectives

- In this chapter, students will learn:
 - What star schemas are and how they are constructed
 - About data analytics, data mining, and predictive analytics
 - About online analytical processing (OLAP)
 - How SQL extensions are used to support OLAP-type data manipulations

Business Intelligence (BI)

- Comprehensive, cohesive, integrated set of tools and processes
 - Captures, collects, integrates, stores, and analyzes data
- Purpose - Generate and present information to support business decision making
- Allows a business to transform:
 - Data into information
 - Information into knowledge
 - Knowledge into wisdom

Figure 13.1 - Business Intelligence Framework



Basic BI Architectural Components

TABLE 13.2

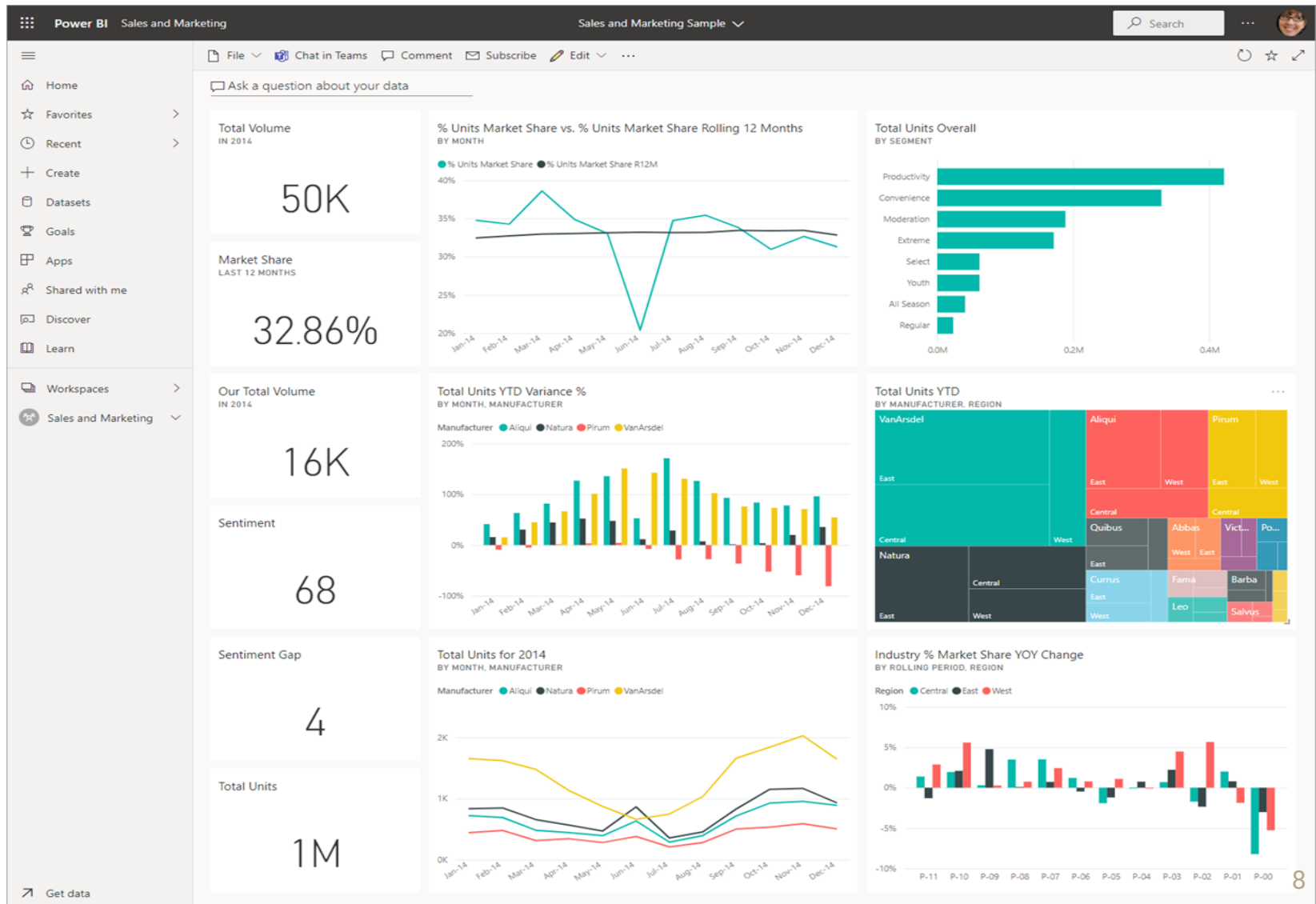
BASIC BI ARCHITECTURAL COMPONENTS

COMPONENT	DESCRIPTION
ETL tools	Data extraction, transformation, and loading (ETL) tools collect, filter, integrate, and aggregate internal and external data to be saved into a data store optimized for decision support.
Data store	The data store is optimized for decision support and is generally represented by a <i>data warehouse</i> or a <i>data mart</i> . The data is stored in structures that are optimized for data analysis and query speed.
Query and reporting	This component performs data selection and retrieval, and it is used by the data analyst to create queries that access the database and create the required reports.
Data visualization	This component presents data to the end user in a variety of meaningful and innovative ways. This tool helps the end user select the most appropriate presentation format, such as summary reports, maps, pie or bar graphs, mixed graphs, and static or interactive dashboards.
Data monitoring and alerting	This component allows real-time monitoring of business activities. The BI system will present concise information in a single integrated view. This integrated view could include specific metrics about the system performance or activities, such as number of orders placed in the last four hours, number of customer complaints by product by month, and total revenue by region. Alerts can be placed on a given metric; once the value of a metric goes below or above a certain baseline, the system will perform a given action, such as emailing shop floor managers, presenting visual alerts, or starting an application.
Data analytics	This component performs data analysis and data-mining tasks using the data in the data store. This tool advises the user as to which data analysis tool to select and how to build a reliable business data model. Business models are generated by special algorithms that identify and enhance the understanding of business situations and problems.

Business Intelligence Tools

- Dashboards and business activity monitoring
 - Dashboards: Shows key business performance indicators in a single integrated view
- Portals: Integrate data using web browser from multiple sources into a single webpage
- Data analysis and reporting tools
- Data-mining tools
- Data warehouses (DW)
- OLAP(**Online analytical processing**) tools and data visualization

Dashboards



Practices to Manage Data

- Master data management (MDM): Collection of concepts, techniques, and processes for identification, definition, and management of data elements
- Governance: Method of government for controlling business health and for consistent decision making
- Key performance indicators (KPI): Numeric or scale-based measurements that assess company's effectiveness in reaching its goals

Practices to Manage Data

- Data visualization: Abstracting data to provide information in a visual format
 - Enhances the user's ability to efficiently comprehend the meaning of the data
- Techniques
 - Pie charts and bar charts
 - Line graphs
 - Scatter plots
 - Gantt charts
 - Heat maps



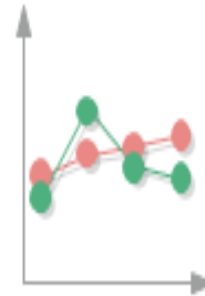
Pie



Bar



Column



Line



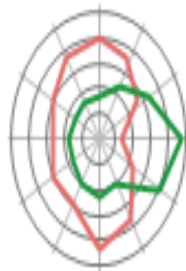
Area



Doughnut



Bubble Chart



Spider and Radar



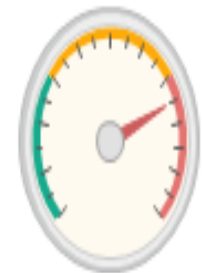
Scatter



Comparison Chart



Stacked bar chart



Gauges

Reporting Styles of a Modern BI System

Advanced reporting

Monitoring and
alerting

Advanced data
analytics

Business Intelligence Benefits

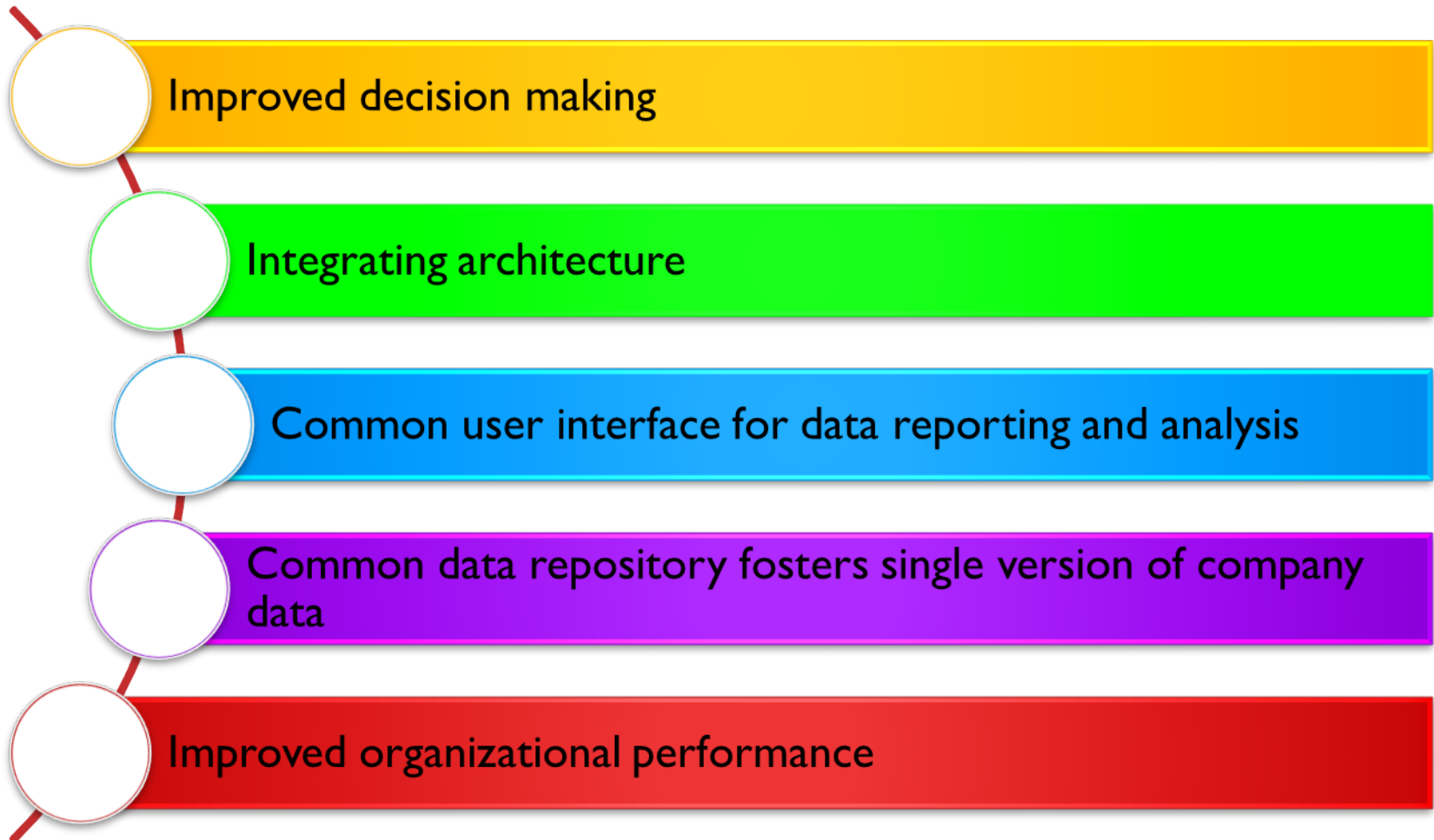


Table 13.4 - Business Intelligence Evolution

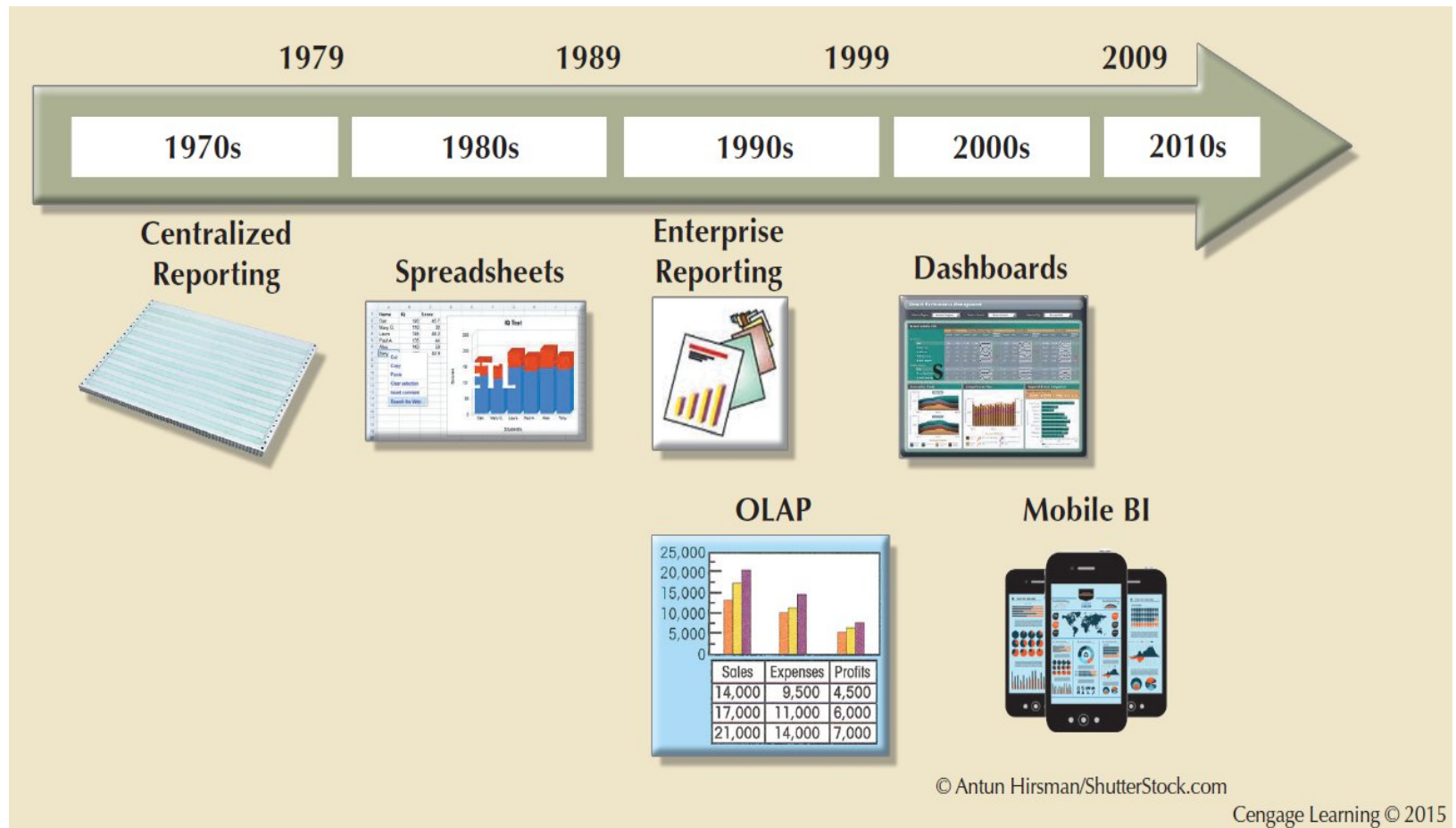
BUSINESS INTELLIGENCE EVOLUTION

SYSTEM TYPE	DATA SOURCE	DATA EXTRACTION/ INTEGRATION PROCESS	DATA STORE	END-USER QUERY TOOL	END USER PRESENTATION TOOL
Traditional mainframe-based online transaction processing (OLTP)	Operational data	None Reports read and summarized data directly from operational data	None Temporary files used for reporting purposes	Very basic Predefined reporting formats Basic sorting, totaling, and averaging	Very basic Menu-driven, predefined reports, text and numbers only
Managerial information system (MIS)	Operational data	Basic extraction and aggregation Read, filter, and summarize operational data into intermediate data store	Lightly aggregated data in RDBMS	Same as above, in addition to some ad hoc reporting using SQL	Same as above, in addition to some ad hoc columnar report definitions
First-generation departmental decision support system (DSS)	Operational data External data	Data extraction and integration process populates DSS data store Run periodically	First DSS database generation Usually RDBMS	Query tool with some analytical capabilities and predefined reports	Spreadsheet style Advanced presentation tools with plotting and graphics capabilities

Table 13.4 - Business Intelligence Evolution

SYSTEM TYPE	DATA SOURCE	DATA EXTRACTION/ INTEGRATION PROCESS	DATA STORE	END-USER QUERY TOOL	END USER PRESENTATION TOOL
First-generation BI	Operational data External data	Advanced data extraction and integration Access diverse data sources, filters, aggregations, classifications, scheduling, and conflict resolution	Data warehouse RDBMS technology Optimized for query purposes Star schema model	Same as above	Same as above, in addition to multidimensional presentation tools with drill-down capabilities
Second-generation BI Online analytical processing (OLAP)	Same as above	Same as above	Data warehouse stores data in MDBMS Cubes with multiple dimensions	Adds support for end-user-based data analytics	Same as above, but uses cubes and multidimensional matrixes; limited by cube size Dashboards Scorecards Portals
Third-generation Mobile, cloud-based, and Big Data	Same as above Includes social media and machine-generated data	Same as above Cloud-based	Same as above Cloud-based Hadoop and NoSQL databases	Advanced analytics Flexible ad hoc interactions via data visualization	Mobile devices: smartphones and tablets

Figure 13.3 - Evolution of BI Information Dissemination Formats



Business Intelligence Technology Trends

Data storage improvements

Business intelligence appliances

Business intelligence as a service

Big Data analytics

Personal analytics

Decision Support Data

- Effectiveness of BI depends on quality of data gathered at operational level
- Operational data
 - Seldom well-suited for decision support tasks
 - Stored in relational database with highly normalized structures
 - Optimized to support transactions representing daily operations

Decision Support Data

- Differ from operational data in:
 - **Time span**: decision support data tends to cover a longer time frame
 - **Granularity**: Decision support data must be presented at different levels of aggregation, from highly summarized to nearly atomic.
 - **Dimensionality**: Operational data focuses on representing individual transactions rather than the effects of the transactions over time

Table 13.5 - Contrasting Operational and Decision Support Data Characteristics

CHARACTERISTIC	OPERATIONAL DATA	DECISION SUPPORT DATA
Data currency	Current operations Real-time data	Historic data Snapshot of company data Time component (week/month/year)
Granularity	Atomic-detailed data	Summarized data
Summarization level	Low; some aggregate yields	High; many aggregation levels
Data model	Highly normalized Mostly relational DBMSs	Non-normalized Complex structures Some relational, but mostly multidimensional DBMSs
Transaction type	Mostly updates	Mostly query
Transaction volumes	High-update volumes	Periodic loads and summary calculations
Transaction speed	Updates are critical	Retrievals are critical
Query activity	Low to medium	High
Query scope	Narrow range	Broad range
Query complexity	Simple to medium	Very complex
Data volumes	Hundreds of gigabytes	Terabytes to petabytes

Decision Support Database Requirements

- Database schema
 - Must support complex, non-normalized data representations
 - Data must be aggregated and summarized
 - Queries must be able to extract multidimensional time slices

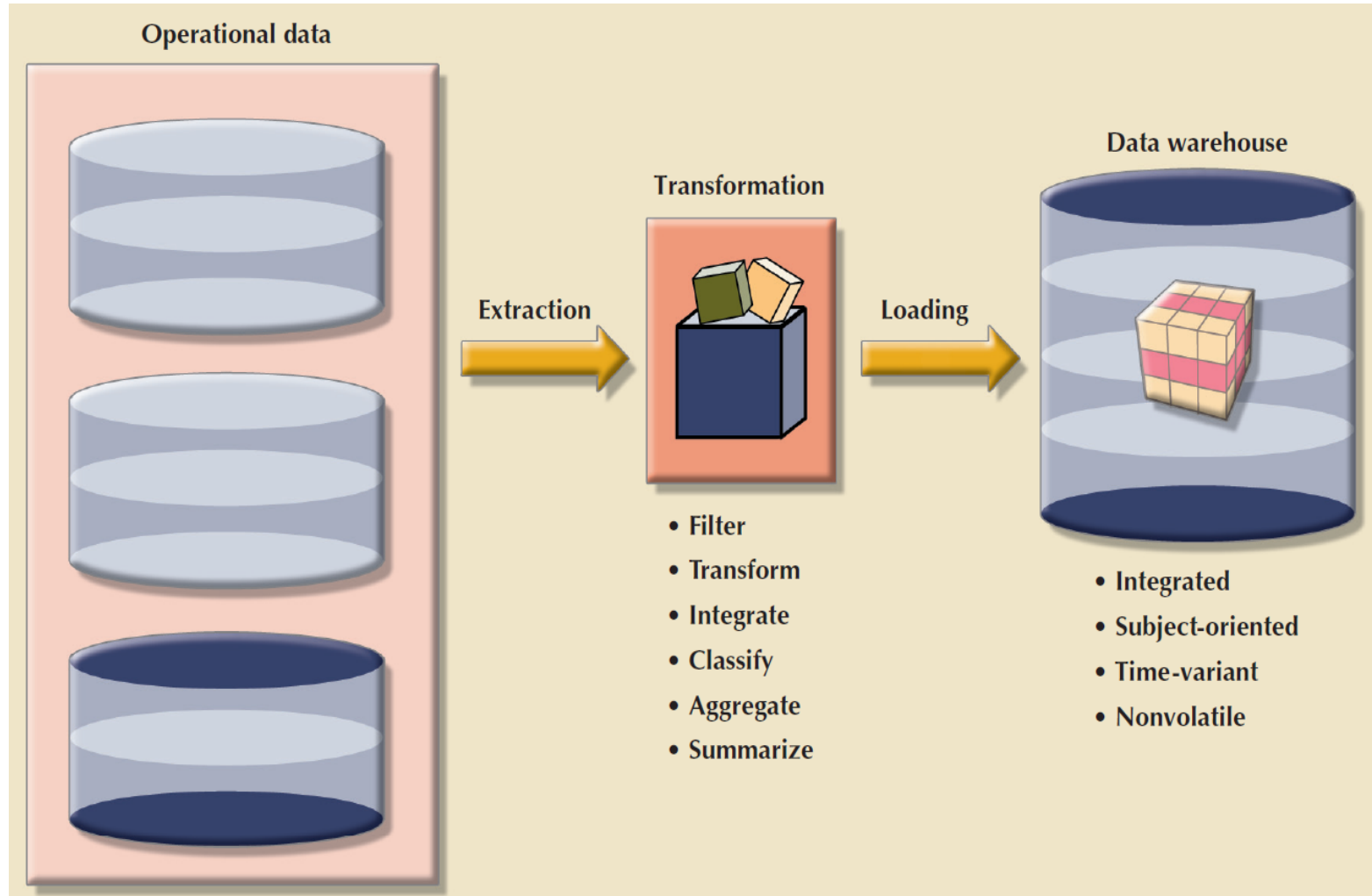
Decision Support Database Requirements

- Data extraction and loading
 - Allow batch and scheduled data extraction
 - Support different data sources and check for inconsistent data or data validation rules
 - Support advanced integration, aggregation, and classification
- Database size should support:
 - Very large databases (VLDBs)
 - Advanced storage technologies
 - Multiple processor technologies

Table 13.8 - Characteristics of Data Warehouse Data and Operational Database Data

CHARACTERISTIC	OPERATIONAL DATABASE DATA	DATA WAREHOUSE DATA
Integrated	Similar data can have different representations or meanings. For example, Social Security numbers may be stored as ###-##-#### or as #####, and a given condition may be labeled as T/F or 0/1 or Y/N. A sales value may be shown in thousands or in millions.	Provide a unified view of all data elements with a common definition and representation for all business units.
Subject-oriented	Data are stored with a functional, or process, orientation. For example, data may be stored for invoices, payments, and credit amounts.	Data are stored with a subject orientation that facilitates multiple views of the data and decision making. For example, sales may be recorded by product, division, manager, or region.
Time-variant	Data are recorded as current transactions. For example, the sales data may be the sale of a product on a given date, such as \$342.78 on 12-MAY-2014.	Data are recorded with a historical perspective in mind. Therefore, a time dimension is added to facilitate data analysis and various time comparisons.
Nonvolatile	Data updates are frequent and common. For example, an inventory amount changes with each sale. Therefore, the data environment is fluid.	Data cannot be changed. Data are added only periodically from historical systems. Once the data are properly stored, no changes are allowed. Therefore, the data environment is relatively static.

Figure 13.5 - The ETL Process



Data Marts

- Small, single-subject data warehouse subset
- Provide decision support to a small group of people
- Benefits over data warehouses
 - Lower cost and shorter implementation time
 - Technologically advanced
 - Inevitable people issues

Table 13.9 - Twelve Rules for a Data Warehouse

RULE NO.	DESCRIPTION
1	The data warehouse and operational environments are separated.
2	The data warehouse data are integrated.
3	The data warehouse contains historical data over a long time.
4	The data warehouse data are snapshot data captured at a given point in time.
5	The data warehouse data are subject oriented.
6	The data warehouse data are mainly read-only with periodic batch updates from operational data. No online updates are allowed.

Table 13.9 - Twelve Rules for a Data Warehouse

RULE NO.	DESCRIPTION
7	The data warehouse development life cycle differs from classical systems development. Data warehouse development is data-driven; the classical approach is process-driven.
8	The data warehouse contains data with several levels of detail: current detail data, old detail data, lightly summarized data, and highly summarized data.
9	The data warehouse environment is characterized by read-only transactions to very large data sets. The operational environment is characterized by numerous update transactions to a few data entities at a time.
10	The data warehouse environment has a system that traces data sources, transformations, and storage.
11	The data warehouse's metadata are a critical component of this environment. The metadata identify and define all data elements. The metadata provide the source, transformation, integration, storage, usage, relationships, and history of each data element.
12	The data warehouse contains a chargeback mechanism for resource usage that enforces optimal use of the data by end users.

Star Schema

- Data-modeling technique
- Maps multidimensional decision support data into a relational database
- Creates the near equivalent of multidimensional database schema from existing relational database
- Yields an easily implemented model for multidimensional data analysis

Components of Star Schemas

Facts

- Numeric values that represent a specific business aspect

Dimensions

- Qualifying characteristics that provide additional perspectives to a given fact

Attributes

- Used to search, filter, and classify facts
- **Slice and dice:** Ability to focus on slices of the data cube for more detailed analysis

Attribute hierarchy

- Provides a top-down data organization

Facts

- Are numeric measurements (values) that represent a specific business aspect or activity.
- For example, sales figures are numeric measurements that represent product and service sales.
- Facts commonly used in business data analysis are units, costs, prices, and revenues.
- The fact table contains facts that are linked through their dimensions.
- Facts are normally stored in a fact table that is the centre of the star schema.
- The fact table is updated periodically with data from operational databases.

Dimensions

- Dimensions are qualifying characteristics that provide additional perspectives to a given fact.
- For instance, sales might be compared by product from region to region and from one time period to the next.
- The kind of problem typically addressed by a BI system might be to compare the sales of unit X by region for the first quarters of 2006 through 2016.
- In that example, sales have product, location, and time dimensions.
- Such dimensions are normally stored in dimension tables.

Attributes

- Each dimension table contains attributes.
- Attributes are often used to search, filter, or classify facts.
- Dimensions provide descriptive characteristics about the facts through their attributes.
- Therefore, the data warehouse designer must define common business attributes that will be used by the data analyst to narrow a search, group information, or
- describe dimensions.
- Using a sales example, some possible attributes for each dimension are illustrated in Table 13.10

Attributes

TABLE 13.10

POSSIBLE ATTRIBUTES FOR SALES DIMENSIONS

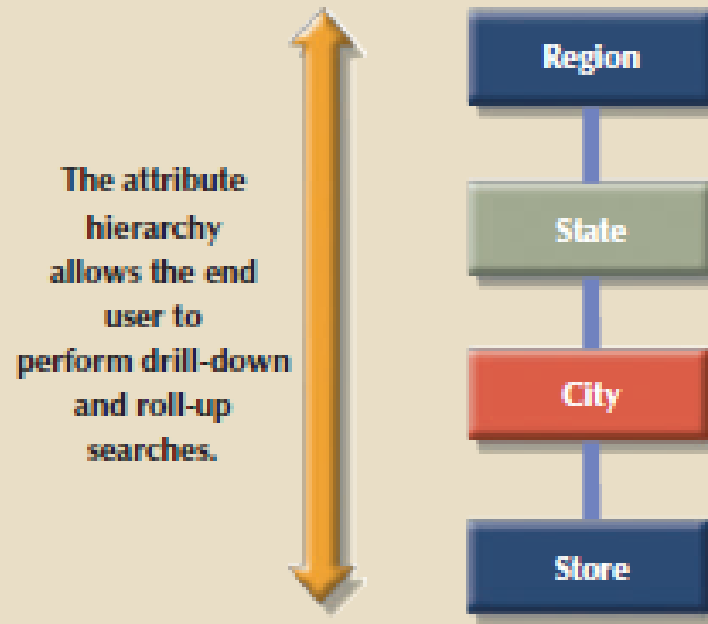
DIMENSION NAME	DESCRIPTION	POSSIBLE ATTRIBUTES
Location	Anything that provides a description of the location—for example, Nashville, Store 101, South Region, and TN	Region, state, city, store, and so on
Product	Anything that provides a description of the product sold—for example, hair care product, shampoo, Natural Essence brand, 5.5-oz. bottle, and blue liquid	Product type, product ID, brand, package, presentation, color, size, and so on
Time	Anything that provides a time frame for the sales fact—for example, the year 2018, the month of July, the date 07/29/2018, and the time 4:46 p.m.	Year, quarter, month, week, day, time of day, and so on

Attribute Hierarchies

- Attributes within dimensions can be ordered in a well-defined attribute hierarchy.
- The attribute hierarchy provides a top-down data organization that is used for two main purposes:
- Aggregation and drill-down/roll-up data analysis.
- For example, Figure 13.8 shows how the location dimension attributes can be organized in a hierarchy by region, state, city, and store.

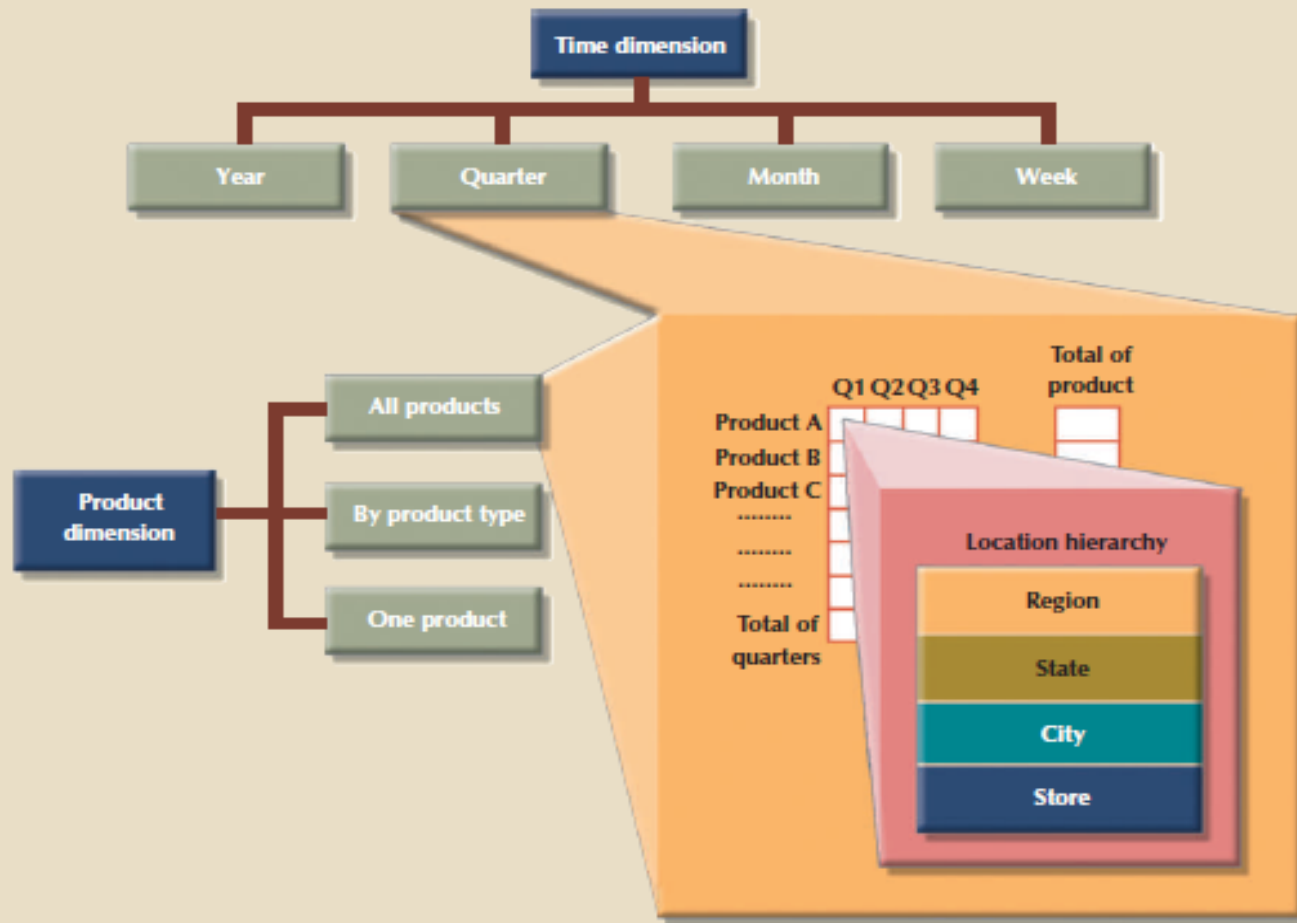
Attribute Hierarchies

FIGURE 13.8 LOCATION ATTRIBUTE HIERARCHY



Attribute Hierarchies

FIGURE 13.9 ATTRIBUTE HIERARCHIES IN MULTIDIMENSIONAL ANALYSIS



Star Schema Representation

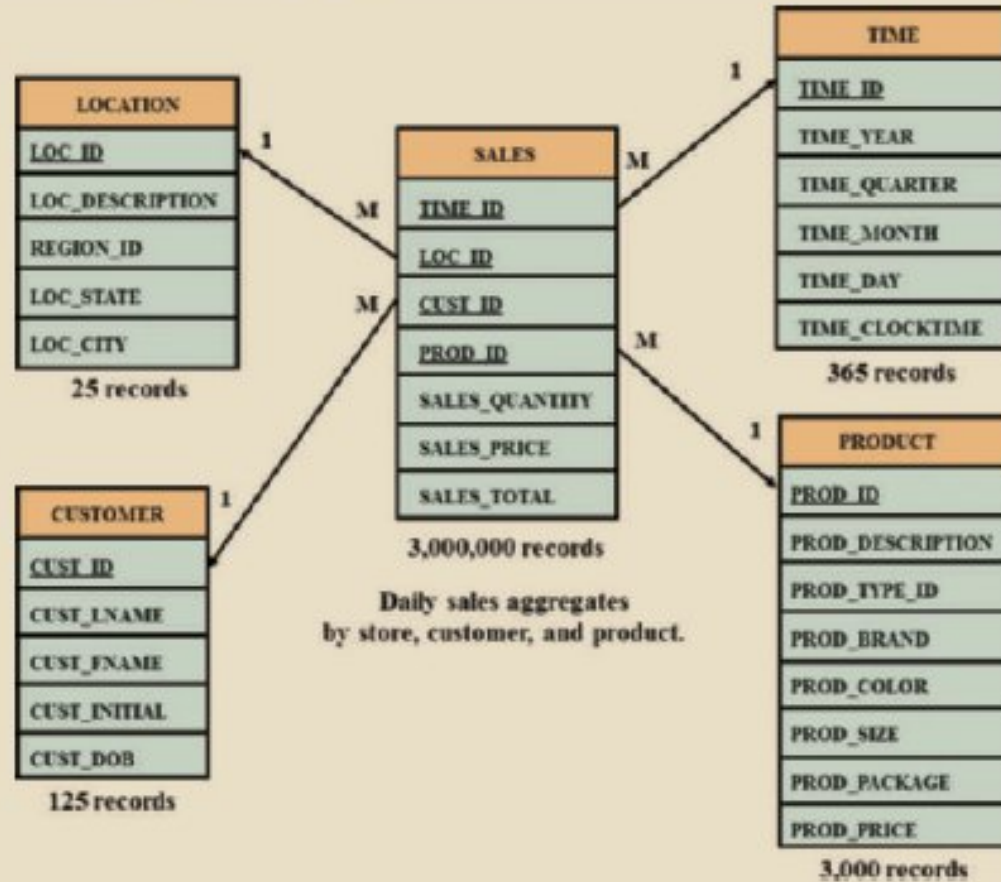
- Facts and dimensions represented by physical tables in data warehouse database
- Many-to-one (M:1) relationship between fact table and each dimension table
- In other words, many fact rows are related to each dimension row
- Fact and dimension tables
 - Related by foreign keys
 - Subject to primary and foreign key constraints

Star Schema Representation

- Primary key of a fact table
 - Is a composite primary key because the fact table is related to many dimension tables
 - Always formed by combining the foreign keys pointing to the related dimension tables

Star Schema Representation

FIGURE 13.10 STAR SCHEMA FOR SALES



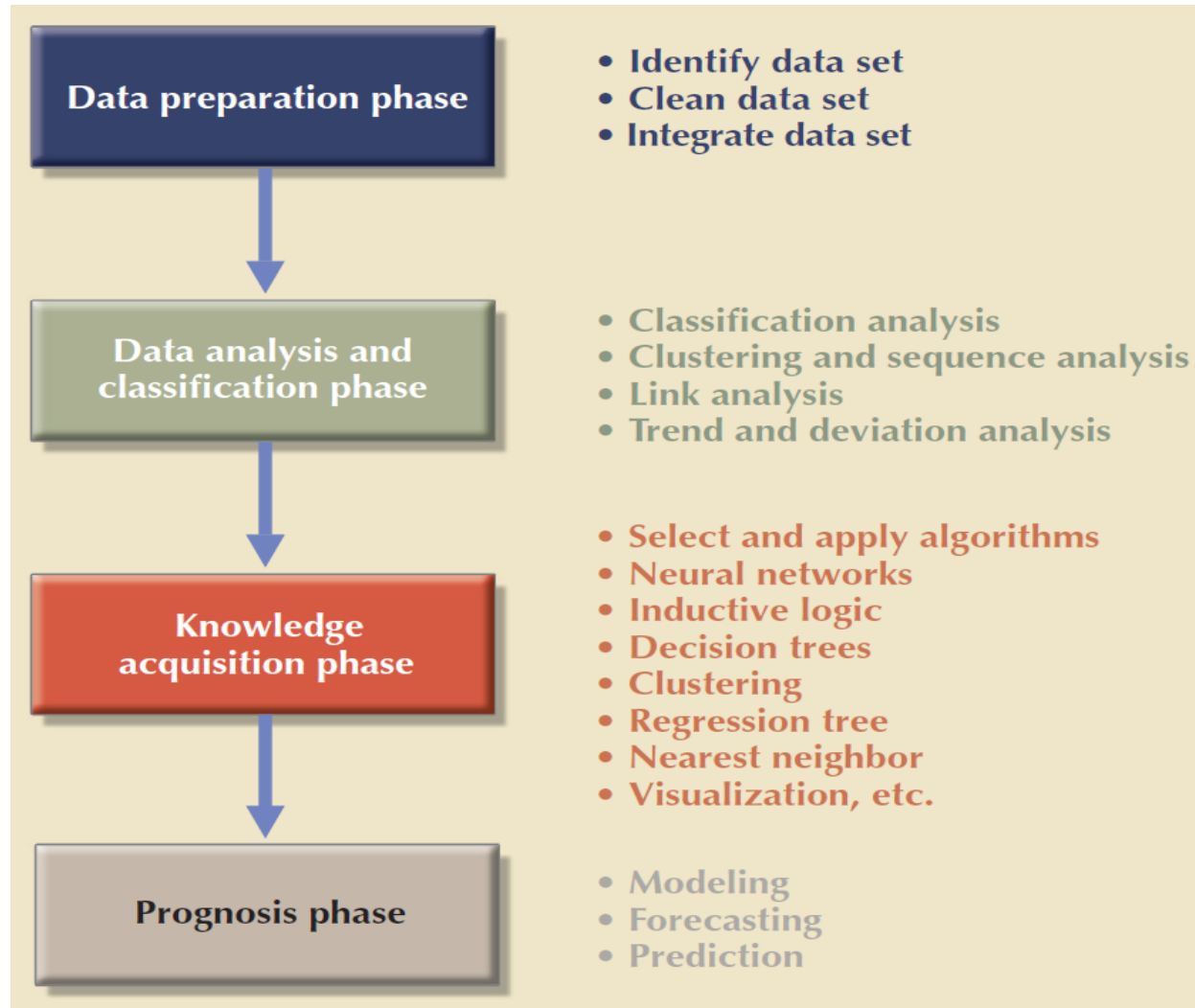
Data Analytics

- Encompasses a wide range of mathematical, statistical, and modeling techniques to extract knowledge from data
 - Subset of BI functionality
- Classification of tools
 - Explanatory analytics: Focuses on discovering and explaining data characteristics and relationships based on existing data
 - Predictive analytics: Focuses on predicting future outcomes with a high degree of accuracy

Data Mining

- Analyzing massive amounts of data to:
 - Uncover hidden trends, patterns, and relationships
 - Form computer models to stimulate and explain the findings
 - Use the models to support business decision making
- Run in two modes
 - Guided
 - Automated

Figure 13. 16 - Data-Mining Phases



Predictive Analytics

- Employs mathematical and statistical algorithms, neural networks, artificial intelligence, and other advanced modeling tools
- Creates actionable predictive models based on available data
 - Next logical step after data mining
- Adds value to an organization
 - Helps optimize the existing processes
 - Identify hidden problems
 - Anticipate future problems or opportunities